IoT-Based Wearable Device for Alzheimer’s Patients | Smart Health Monitoring Project

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In a world where the technology is everywhere, what if a small wearable device can save a life? What if your parents or grandparents never feel alone when someone is not around? Hi, this is Krishna Seshkant Gunkar, Embedded Systems Engineer at Trainsversity. Today I bring you something, it's not just a project, it is an emotion, where Alzheimer's disease. Alzheimer's disease is a disease which occurs a brain neurological disorder.

When a people at the age of 65, they are suffering from Alzheimer's disease, it is like forgetting a recent events and forgetting how to take a medicine and how to take a food. So this is what we can build a wearable device. So today we are going togetherly build a project, IoT wearable device for Alzheimer's disease.

Imagination is an innovation. Imagine this, if your grandfather go for a walk, he doesn't come back, but he forgot the way, he forgot he has the diabetes, he forgot how to take a medicine because of Alzheimer's disease. This Alzheimer's disease is not only steals the memories, it puts a life in a risk.

What if we can build a device that can monitor health like parameters, BP, SpO2 and temperature, humidity. So monitoring this all the parameters, send data to the cloud. So this is all about IoT based wearable device.

Now I will explain how you can build. Now let's talk about how it works and what is inside. So before going to do any project, so first we need to consider a three parameters such as first one is hardware implementation, second one is code debugging and third one is prototyping.

So in a hardware implementation, so first we need to configure what all are the components is to be required to do the project. So our all components is depends on our project objectives. So in this our project first objective is we are going to measure a health parameter such as body temperature, humidity, SpO2 and BP.

So this is our first objective. Second objective is our, so after measuring all the health parameters, then we need to push a data to the cloud using ThinkSpeak. And third objective is we are giving a continuous reminder to the patients using RTC driver.

So after that, once the RTC model is being initialized with the CMOS battery, it will give like real time date and a time. So based on that, we'll keep every three hour alarm such as time for lunch and time for medicine. So after third objective is our real time.

Once the patient is fall, then we need to detect the fall detection and give the emergency alert to the caretakers. So this is all about our major implementations and our objective. So let's start

the brain of the system is ESP32 Deo model.

So this is the ESP32 Deo model is a powerful microcontroller that connects to the sensors and upload data to the cloud. After that, this is a body temperature and humidity using DHT11 sensor. Then we are going to monitor heart rate and oxygen level using a MAX30100 sensors and movement for detection will be measured by accelerometer ADXL335 and we are sending a real time reminders using RTC model.

So RTC model is stands to be a real time clock. So we are using specific real time clock is DS3231. So we are using GSM900 so used to send a message to the caretakers.

So this is all about our project hardware implementations and components required. So now for fall detection, we are using Arduino UNO as a microcontroller. Similarly, followed by GSM800C so and ADXL335 is an accelerometer sensor.

So here accelerometer sensor as soon as we detect, then it will impact is been detected. Then it is send a signal to the controller. Then as soon as signal received by the controller, it send again signal to the GSM.

So when a GSM is received a signal, then it will initiate the call to the user stating that your patient is fall down. So kindly take care. So this is all about a fall detection.

It is also one of the objective. So after selecting a component, then we begin with the connection. So connection is based on connection table.

So here components are one side and ESP32 pin diagram means one side. So here first we'll start with the DST11 sensor. So DST11 sensor is an sensor which measures the health parameters such as a body temperature and humidity.

So DST pin number out is going to connect GPIO13 of ESP32 and VCC will be going to connect 3.3 volt and ground pin will going to connect ground of the ESP32. Similarly MAX30100 pulse oximeter is an a sensor which going to measure the health parameter such as SpO2 and BP. So here SDA will going to connect GPIO21, SCL will going to be connect GPIO22, VIN pin will going to connect a 3.3 volt and ground pin will going to be connect ground.

Similarly OLED display, again OLED display is also work on I2C. So again here SDA will going to connect GPIO21, SCL will going to be connect GPIO22, VCC will going to be connect 3.3 volt and ground pin will going to be connect ground of ESP32. Similarly again RTC module, RTC module is also work on I2C protocol.

Again here same connection as OLED and MAX30100 sensor. SDA pin will going to be connect GPIO21, SCL pin will going to be connect GPIO22, VCC is 3.3 volt, ground is again ground. So now the question is how we all going to be connect SDA and SCL in one board.

So we need to take a common connection from ESP32 in the breadboard. So SDA will take one

common and SCL also will take one common using breadboard. Again as a buzzer for indication purpose we are going to connect GPIO26 and ground will be ground.

Wi-Fi is inbuilt, again thingspeak also we need to create one account on thingspeak. So data can be uploaded using thingspeak. So here this all the connection based on the connection table.

So after completion of the connection diagram with the help of connection table then after we need to begin with the code debug. So here I return the code for the entire program and entire project. So here I included all the libraries which all are necessary for this project.

Then here after I included my SSID and password. So SSID and password is nothing but your Wi-Fi credentials. Similarly here thingspeak API key and thingspeak.com like thingspeak.com and after that we will begin with an initialization.

So here all the initializations will come. Then once initialization is been done here we will set the timers like 15 and 30. So lunch alarm and medicine alarm.

Similarly after the alarm is been completed then we will keep some buzzer for indication purpose. So this is all about our code. So how a program is start and how the temperature is going to be measured and SpO2BP.

So after measuring this all the parameters we send data to the cloud. So with the help of this program first we need to set the alarm for this project. So here now as per the time schedule 15, 30, 32 and 15, 34 go for upload.

It's take a time for compiling. So during the compilation you need to hold the boot button. So after compilation code is uploading into the microcontroller.

As soon as code is been upload then just click on a reset button. So Wi-Fi is connected. This is all IP address.

Then RTC clock is been started. So there your time and date and will be measured and here a temperature BP and SpO2. Now you can check all the data on a ThinkSpeak.

So in this ThinkSpeak you need to create a channel. So after creating the channel so you need to visit your channel. Then here you need to create the field like I created temperature, humidity.

Field 1 is temperature, field 2 is humidity and field 3 is BP and SpO2. So here all the four parameters you have to measure. Then once the parameter is been measured then go for data import.

So here click on export and download your sheets. So as soon as your sheet is been downloaded then you can visible your data. Like today how much we measure each and everything will be followed.

So date wise here are the date and this is the entry how much entry you did till today. And this is the field 1, field 2, field 3, field 5 like that. You need to get the overall data.

So after measuring this overall data we are giving access to the doctors. So based on this all the analysis and data doctor will give the medicine to the patients. So this is what about our project and after every alarm it will show the time for lunch and time for medicine.

So now you are seeing time for lunch. So here after 15.32 whatever the alarm we are initialized in the code so that is been triggered. So now this is for a lunch time.

After that after 15.34 we given the alarm for the medicine. It is the time for lunch with buzzer. So after second alarm 15.34 it will print time for medicine.

So this is both alarm we are giving for the patient stating that first one is for lunch reminder and second one is for medicine reminder. So this is how RTC model will work. We use the reminder purpose for the patient.

So after measuring a health parameter we start with a fall detection. So in the fall detection we are using Arduino Uno is a controller and GSM for transmitting a message to the caregivers. And this is an ADXL 3.5 accelerometer.

So in this project I have included code. So this code is all about detecting the fall detection. First you need to initialization the pin of GSM.

So here I return 2 and 3 pin for RX and TX. While making connection you need to ensure that RX pin will going to be connect TX of Arduino and TX pin will going to be connect RX pin of the Arduino. Similarly this is the connection for ADXL 3.5 accelerometer.

So A1 pin of ADXL you need to connect X pin you need to connect A1 Y pin you need to connect A2 Z pin you need to connect A3. So after the connection this is all the logic. So with the baud rate of 9600.

So here as soon as this all the command AT commands will be initialized. Then we are giving a impact threshold value. So impact is been nothing but whatever the sensor data is been calculated with the help of ADXL.

So we are giving some impact threshold value. So as soon as it will threshold will process then the call is been initiated to the caregiver. So here when I tap this sensor in the command in the serial window it will show how much impact is been detected.

So we are giving 150 is a threshold value. So it crosses the threshold value then after that it will take a call is been initiated to the caregiver. So here when I tap here impact is been detected with the 150 magnitude.

Making a call. Call initiated successfully. This is a call.

So this is the fault detection all about. So this is how our entire project is going to be work with the help of ESP32 and Arduino UNO. So using a GSM implementation.

So this is all about our entire project. This project is not just about the electronics. It is about giving hope to the families who lives fear in every day.

And it's about giving independence to the patients. And peace of mind for the caregivers. This is the power of IoT.

Thanks for watching. Hit like and subscribe this channel for more embedded projects.

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